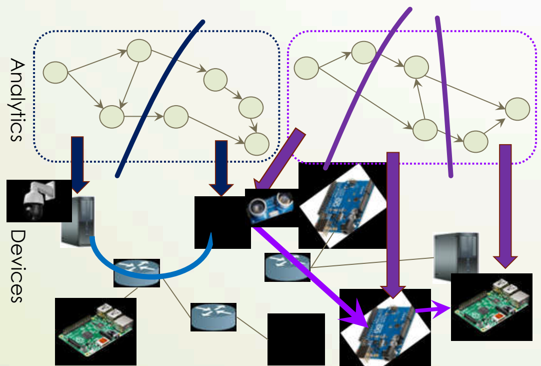
SSE Runs Fast Enough While Achieve Lower Analytics Latency

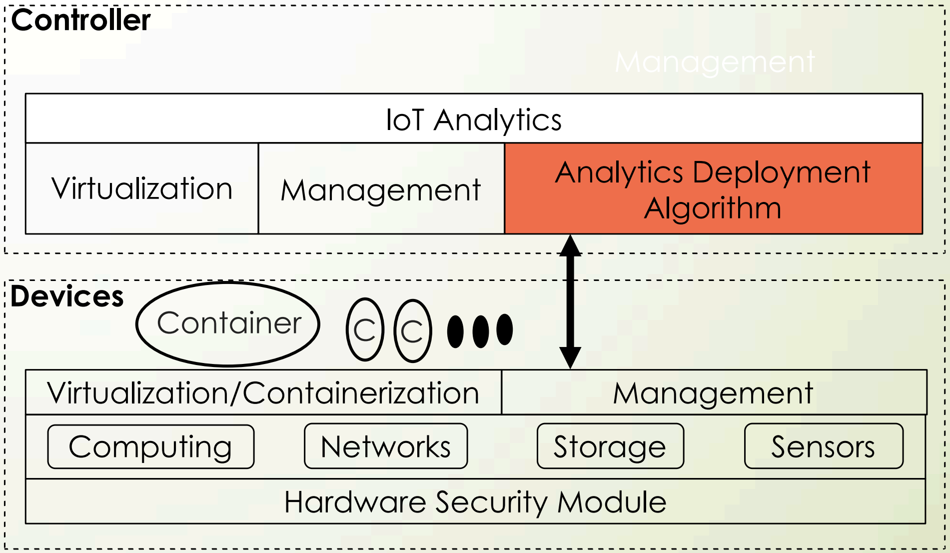
No. Devices	10	25	50	75	100	150
SSE	1.07	2.41	5.36	8.88	9.78	13.52
ODP	2.37	22.66	101.93	382.83	>500	>500
FCP	0.07	0.10	0.21	0.27	0.31	0.46

Computation overhead
- Container: 5%
- Distributed TensorFlow: 10%

Summary: Analytics Deployment Problem

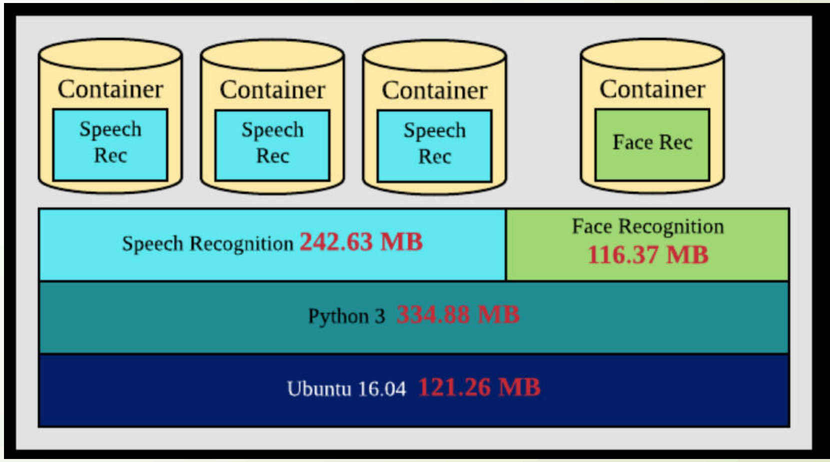
- We solve the problem in three steps
  - Resource consumption modeling
  - Analytics decomposition
  - Polynomial algorithms
- Solved two formulations with APX/SSE algorithms
  - APX: $1/|U|$ factor
  - SSE : 46+% more analytics





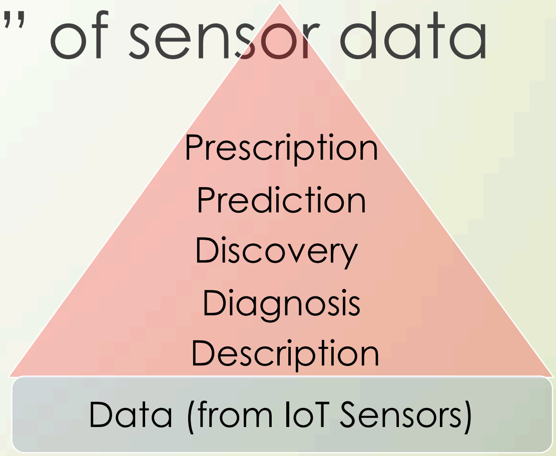
Current/Future Extensions

- Efficiently retrieve (potentially large) container images
  - Preload images (but where, when?)
- Lighter-weight environments?
  - Sandbox and unikernel
- Disseminate results to (nearby) end users
  - Distributed pub-sub brokers
  - Data/information storage
- Better integration with Trigger-action or IFTTT (if this, then that) rules



Fundamental Problem in Cloud-to-Things Continuum: Gap Between Data and Knowledge

- Thus far, we use "analytics throughput", say frame-per-second, as the QoS metric
  - Rate-Quality curves as in the multimedia literature
  - Do we really know which analytics are needed before hand?
- How can we determine the "values" of sensor data before collecting them?
  - Higher in the knowledge hierarchies?
  - Probably a function of context?
  - Or even depends on roles?



Questions

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